Ad Optimization Agent – Group Task Distribution (6 Members)

Member 1 – Agent Design & System Prompt - Chelsey

- Write the 1-page design document.
- Explain the agent's role, inputs, outputs, rules, privacy, and guardrails.
- Create the system prompt for the agent.

Member 2 – Dataset Creation- Chelsey

- Create a CSV dataset with 14–30 days of advertising data.
- Include Search, Social, and Display channels.
- Include date, spend, impressions, clicks, and conversions.
- Share the completed dataset with the coding member.

Member 3 – Agent Coding & Budget Allocation - Derek

- Write the Python script using Google Colab.
- Read and analyze the CSV dataset.
- Calculate conversion rates and CPA.
- Develop the logic to recommend daily budget allocations.
- Apply budget limits and log decisions with explanations.

Member 4 – Evaluation & Testing - Devin

- Test the agent using the dataset.
- Calculate total conversions, CTR, and average CPA.
- Evaluate the agent's performance and document the results.
- Work with Member 3 to verify the code works correctly.

Member 5 – README & Scaling Note - Hanzon

- Write instructions explaining how to run the code.
- Document assumptions and summarize results.
- Write the half-page scaling note covering cloud deployment, monitoring, retries, and cost limits.

Member 6 – GitHub, Presentation & Submission

- Organize the GitHub repository and upload project files.
- Prepare the 2–3 minute presentation slides.
- Collect screenshots or outputs showing successful code execution.
- Coordinate the final submission and ensure all requirements are completed.

1. Agent Role and Purpose

The purpose of our Ad Optimization Agent is to help businesses make better decisions about where to spend their advertising budget. Businesses often advertise across multiple platforms, but not every channel delivers the same results. Our agent analyzes the performance of three channels—Search, Social, and Display and recommends how to distribute the daily budget based on which channels are generating the most conversions at a reasonable cost. The goal is to improve advertising performance without increasing overall spending.

2. Inputs and Outputs

The agent uses a CSV dataset containing 14–30 days of advertising performance data. This includes the date, channel, spending, impressions, clicks, and conversions.

Using this information, the agent calculates performance metrics and recommends a budget for each channel for the following day. Each recommendation includes an explanation of why the budget should increase, decrease, or remain the same. These decisions are also recorded so we can track how the agent responds to changes in performance.

3. How the Agent Works

The agent begins with an equal budget allocation across all three channels. As it processes the advertising data, it calculates conversion rate and cost per acquisition (CPA) to compare performance.

For instance, if Search consistently generates more conversions at a lower cost than Social or Display, the agent may recommend increasing its budget. However, rather than shifting too much money at once, it makes gradual adjustments. This allows the agent to take advantage of stronger-performing channels while continuing to collect data from the others.

The agent repeats this process as new performance data becomes available, allowing future recommendations to reflect recent results.

4. Rules and Guardrails

To prevent unnecessary spending or drastic budget changes, we established several rules:

- The total allocation must remain within the daily advertising budget.
- Budget adjustments cannot exceed 20% per channel in a single day.
- Each channel must receive at least 10% of the total budget.
- The agent only analyzes advertising metrics and does not require customers' personal information.
- Existing advertisements and brand messaging remain unchanged.
- Every recommendation must include a reason for the adjustment.
- If the available data is insufficient, the agent maintains the current allocation rather than making an unsupported recommendation.

5. Evaluation Metrics

We will evaluate the agent using total conversions, conversion rate, and average CPA. We will also compare its performance against a baseline where the budget is divided equally across all three channels. This will help us assess whether the agent's recommendations could lead to better advertising efficiency.

Since our project uses simulated data, the results will demonstrate how the agent makes decisions rather than guarantee improvements in real advertising campaigns.

6. System Prompt

You are an Ad Optimization Agent responsible for analyzing advertising performance and recommending daily budget allocations across Search, Social, and Display.

Review the provided CSV data and calculate conversion rates and CPA for each channel. Use these metrics to identify stronger performing channels and recommend adjustments that prioritize conversions while maintaining cost efficiency.

Follow these requirements:

1. Keep the combined allocations within the total daily budget.
2. Limit increases or decreases to 20% of each channel's previous daily allocation.
3. Maintain a minimum of 10% of the total budget for every channel.
4. Use only aggregated advertising data and protect customer privacy.
5. Do not modify advertising content or brand messaging.
6. Provide a clear explanation for every budget recommendation.
7. Maintain the existing allocation when the data is insufficient or unreliable.

Present the recommended budget for each channel, the relevant performance metrics, and the reasoning behind each adjustment.

Google Colab link:
https://colab.research.google.com/drive/1mgfbZIdcwfSIVqSriLGFnvVVQTz4gZYH?usp=sharing
Dataset and output files:
https://drive.google.com/file/d/1fFbm-_Lz7mt6G4ZNPEuE04LQ9f6xMmTM/view?usp=drive_link

README

Ad Optimization Agent

Project Purpose

This project uses advertising data to recommend how a business should divide its daily budget among Search, Social, and Display advertising.
The agent looks at recent spending, impressions, clicks, and conversions. It then recommends a budget for the next day. It also explains whether each channel should be increased, decreased, or kept the same.

Files

- advertising_data.csv - Cleaned copy of the uploaded advertising data.
- channel_metrics.csv - Overall performance for each channel.
- decision_log.csv - Daily recommendations and explanations.
- next_day_budget.csv - The most recent budget recommendation.
- Member3_Results.zip - ZIP file containing the output files. For the final submission, this can be renamed to Ad_Optimization_Agent_Results.zip.

How to Run the Code

1. Open the Google Colab notebook.
2. Run the code cell.
3. When Colab asks for a file, upload the advertising Excel or CSV file.
4. Make sure the file contains these columns:
   date, channel, spend, impressions, clicks, conversions
5. Wait for the code to finish.
6. Review the historical channel performance and the next-day budget recommendation.
7. Colab will download a ZIP file containing the results.

What the Results Mean

- **CTR (%)** shows the percentage of impressions that became clicks.
- **Conversion Rate (%)** shows the percentage of clicks that became conversions.
- **CPA ($)** shows the average cost to get one conversion. A lower CPA is usually better.
- **Recommended Budget** shows how much each channel should receive the next day.
- **Action** shows whether the budget should increase, decrease, or stay the same.
- **The reason** explains why the recommendation was made.

The agent prioritizes channels that produce more conversions for each dollar spent. If there is not enough reliable data, it keeps the previous budget instead of making a risky change.

Assumptions

- The data is simulated advertising data, not guaranteed real-world performance.
- The file contains one row for each date and channel combination.
- The available channels are Search, Social, and Display.
- The daily budget is set to $300 in the code.
- The starting budget is divided as equally as possible among the three channels.
- Each channel must receive at least 10% of the daily budget.
- A channel's budget cannot change by more than 20% in one day.
- Conversions cannot be greater than clicks, and clicks cannot be greater than impressions.
- CPA is undefined when a channel has zero conversions.
- The code uses the most recent seven days of data when making a decision.
- At least three days of data are required for each channel before the agent changes the allocation.
- The recommendations are simulated. They do not prove what would happen after actually spending the new budgets.

Results Summary

After the notebook runs, use channel_metrics.csv to report the total spend, impressions, clicks, conversions, CTR, conversion rate, and CPA for Search, Social, and Display. Use next_day_budget.csv to report the latest recommended budget.

The main result is a daily budget recommendation that stays within the $300 limit, gives every channel at least 10% of the budget, and limits daily changes to 20%. The decision log also gives a reason for every recommendation.

Limitations

The agent does not use revenue, profit, audience quality, seasonality, or ad creative information. It only uses the columns in the uploaded file. Because the data is simulated, the recommendations should be treated as a demonstration of the agent's decision process, not as a promise of better advertising results.

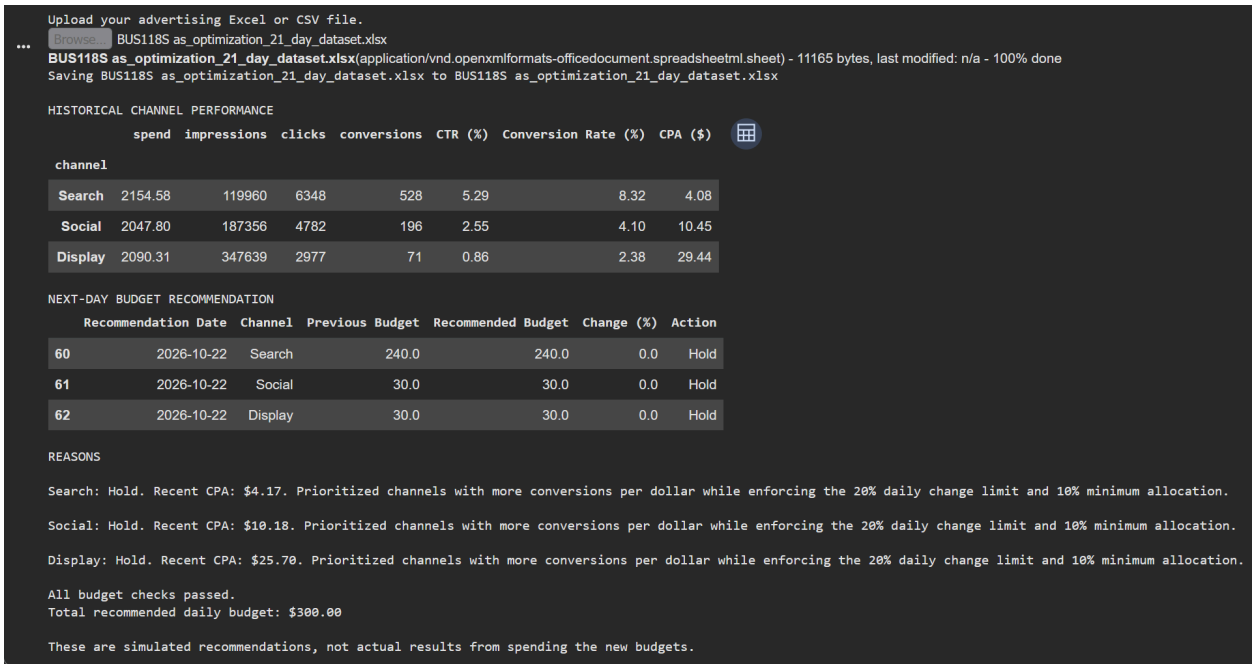
SCALING

The current agent runs in Google Colab and works well for a small class project. To use it for a real business, it could be moved to a cloud service such as Google Cloud Run, AWS Lambda, or Azure Functions. A scheduled job could collect new advertising data each day, run the agent, and save the recommendations in cloud storage or a database.

The system should be monitored after every run. Monitoring should check whether the data arrived, whether all three channels are present, whether the budget equals $300, and whether the 10% minimum and 20% change rules passed. The system should also record the run time, input file, recommendation, errors, and CPA or conversion changes. Alerts could be sent if a file is missing, a channel has no data, or a recommendation fails a rule.
Cloud jobs can sometimes fail because of a temporary network or service problem. The system should retry a failed run a small number of times, such as two or three attempts, with a short wait between attempts. Retries should not create duplicate recommendations. Each run should have a unique ID, and the system should save a result only after all checks pass. If the job still fails, the safest choice is to keep the previous budget and notify a human.

Cost limits are also important. The process should run once per day, use a small cloud service, limit the amount of data processed, and automatically stop or alert an administrator if cloud spending reaches a set limit. The agent should recommend budgets only; a human should approve automatic changes before they are sent to advertising platforms. This keeps the system affordable, easier to monitor, and safer for the business.

RESULTS



Upload your advertising Excel or CSV file.

Browse... BUS118S as_optimization_21_day_dataset.xlsx

BUS118S as_optimization_21_day_dataset.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 11165 bytes, last modified: n/a - 100% done

Saving BUS118S as_optimization_21_day_dataset.xlsx to BUS118S as_optimization_21_day_dataset.xlsx

HISTORICAL CHANNEL PERFORMANCE

channel	spend	impressions	clicks	conversions	CTR (%)	Conversion Rate (%)	CPA ($)
Search	2154.58	119960	6348	528	5.29	8.32	4.08
Social	2047.80	187356	4782	196	2.55	4.10	10.45
Display	2090.31	347639	2977	71	0.86	2.38	29.44

NEXT-DAY BUDGET RECOMMENDATION

	Recommendation Date	Channel	Previous Budget	Recommended Budget	Change (%)	Action
60	2026-10-22	Search	240.0	240.0	0.0	Hold
61	2026-10-22	Social	30.0	30.0	0.0	Hold
62	2026-10-22	Display	30.0	30.0	0.0	Hold

REASONS

Search: Hold. Recent CPA: $4.17. Prioritized channels with more conversions per dollar while enforcing the 20% daily change limit and 10% minimum allocation.

Social: Hold. Recent CPA: $10.18. Prioritized channels with more conversions per dollar while enforcing the 20% daily change limit and 10% minimum allocation.

Display: Hold. Recent CPA: $25.70. Prioritized channels with more conversions per dollar while enforcing the 20% daily change limit and 10% minimum allocation.

All budget checks passed.
Total recommended daily budget: $300.00

These are simulated recommendations, not actual results from spending the new budgets.

Total Conversions: 795
Overall CTR: 2.15%
Average CPA: $7.92

The Ad Optimization Agent processed the dataset, calculated advertising performance metrics, and generated a daily budget recommendation that fits our guidelines. Search was the best channel based on the data shown, and the program completed its budget checks and recorded explanations for its decisions.